

Assignment 1: Imitation Learning

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Collaborators: None

1 Behavioral Cloning (10 pt)

1.1 Part 2 (1.5 pt)

Table 1: Mean and standard deviation of expert return over two trajectories, for each environment.

Metric/Env	Ant-v2	Humanoid-v2	Walker2d-v2	Hopper-v2	HalfCheetah-v2
Mean	4713.65	10344.52	5566.85	3772.67	4205.78
Std.	12.20	20.98	9.24	1.95	83.04

1.2 Part 3 (5 pt)

Table 2: BC vs. expert return. List the environment you chose and the hyperparameters (network size, amount of data, training iterations, `eval_batch_size`, etc.) here.

Hyperparameters: `n_layers` = 3, `size` = 128, `learning_rate` = 10^{-3} ,
`num_agent_train_steps_per_iter` = 10,000, `train_batch_size` = 100, `eval_batch_size` = 5000,
`n_iter` = 1, seed 1. Training data: 2 expert trajectories (2000 transitions).

Env	Ant-v2		Humanoid-v2	
Metric	Mean	Std.	Mean	Std.
Expert	4713.65	12.20	10344.52	20.98
BC	4732.75	100.06	288.89	21.48

1.3 Part 4 (3 pt)

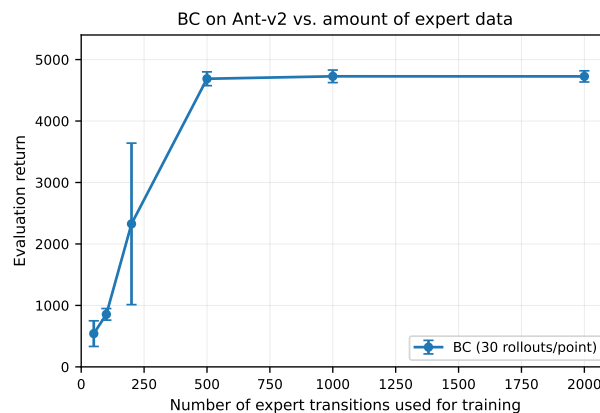


Figure 1: BC on Ant-v2 vs. the number of expert transitions used for training, subsampled from the 2 provided trajectories. Network (3×128), $1r = 10^{-3}$ and 10,000 gradient steps are held fixed; each point is the mean over 30 rollouts (3 seeds $\times$ `eval_batch_size` = 10,000) and the error bars are the std across those rollouts. I expected BC to get gradually worse as I removed data, but it stays at expert level down to 500 transitions and then falls off fast, so my hypothesis was wrong.

1.4 Part 5 (0.5 pt)

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware (2023) introduces ACT (Action Chunking with Transformers) which utilizes behavior cloning. The behaviors are dexterous manipulation skills such as slotting a battery and opening a condiment cup, and the expert data is human teleoperation of a pair of leader arms. The main risk from this homework, compounding error under distribution shift, applies directly since ACT is trained with no DAgger-style interactive relabeling to correct for policy drifting. The paper tries to mitigate this problem by predicting a chunk of k future actions rather than one step at a time, which shortens the effective decision horizon. This reduces how quickly errors accumulate in their experiments.

2 DAgger (5 pt)

2.1 Part 2 (5 pt)

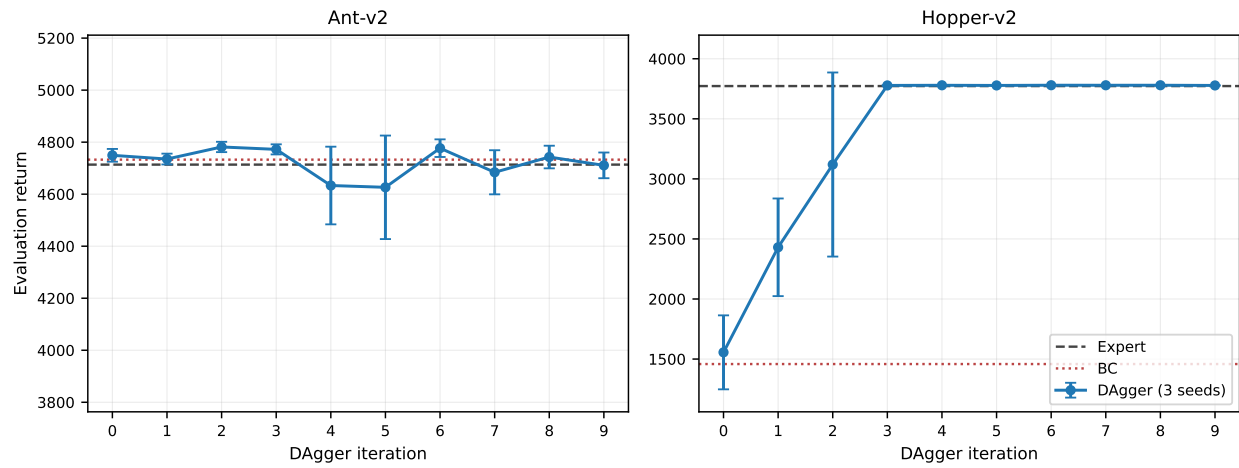


Figure 2: DAgger learning curves (mean return vs. iteration, with error bars). Ant-v2 on the left and Hopper-v2 on the right. Dashed = expert, dotted = BC. Policy: 3×128 MLP, $1r = 10^{-3}$, 10,000 gradient steps/iteration, starting from 2 expert trajectories and adding 1000 relabeled transitions per iteration.